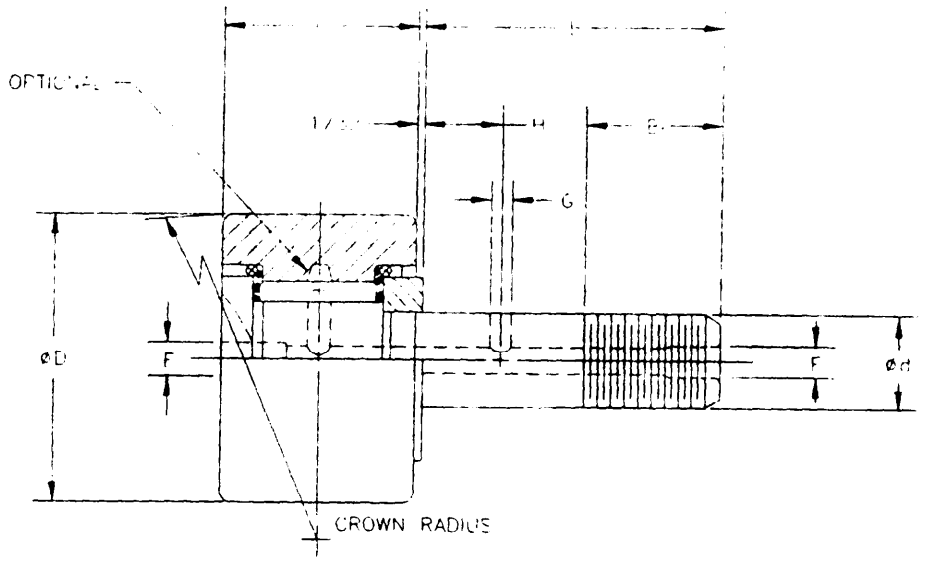
THIS SPECIFICATION IS APPROVED FOR USE BY ALL DEPARTMENTS AND AGENCIES OF THE DEPARTMENT OF DEFENSE.

THE REQUIREMENTS FOR ACQUIRING THE PRODUCT(S) DESCRIBED HEREIN SHALL CONSIST OF THIS SPECIFICATION SHEET AND THE ISSUE OF THE FOLLOWING SPECIFICATION LISTED IN THAT ISSUE OF THE DODISS SPECIFIED IN THE SOLICITATION: MIL-B-3990



(B) TABLE I. DIMENSIONS.

Dash No.	Ød Stud Dia	ØD Dia Outside	C Outer Ring width	B Stud Length	B• Minimum Perfect Thread Length	Fine Threads UNF	Oil Hole ØC Hole Dia	Oil Hole H Hole Center	F Lub Fitting Size	Crown Radius Ref.	Total Radial Clearance Max	Clamping Dia Min.	1/ lbs - in. max.	Capacity As a Track Roller lbf.	Track Capacity 2/	Limit Load Rating lbf	Mass Approx. lbf
-080	.2500	.5000	.375	0.625	0.2500	1/4-28	–	–	0.125•	7.00	0.0023	0.40	10.0	700	508	1000	0.03
-091	.2500	.5625	.375	0.625	0.2500	1/4-28	–	–	0.125•	7.00	0.0023	0.40	10.0	700	–	1000	0.04
-101	.3125	.6250	.438	0.750	0.3125	5/16-24	–	–	0.125•	8.00	0.0023	0.47	40.0	1100	718	1500	0.05
-111	.3125	.6875	.438	0.750	0.3125	5/16-24	–	–	0.125•	8.00	0.0023	0.47	40.0	1100	–	1500	0.06
-121	.4375	.7500	.500	0.875	0.3750	7/16-20	0.094	0.250	0.188	10.00	0.0023	0.61	150.0	2000	1030	2200	0.09
-141	.4375	.8750	.500	0.875	0.3750	7/16-20	0.094	0.250	0.188	10.00	0.0023	0.61	150.0	2000	1160	2600	0.11
-161	.6250	1.0000	.625	1.000	0.5000	5/8-18	0.125	0.250	0.188	12.00	0.0023	0.78	390.0	2900	1480	3900	0.21
-181	.6250	1.1250	.625	1.000	0.5000	5/8-18	0.125	0.250	0.188	12.00	0.0023	0.78	390.0	2900	1670	3900	0.24
-201	.7500	1.2500	.750	1.250	0.6250	3/4-16	0.125	0.312	0.188	14.00	0.0023	0.97	750.0	4100	2330	5500	0.39
-221	.7500	1.3750	.750	1.250	0.6250	3/4-16	0.125	0.312	0.188	14.00	0.0023	0.97	750.0	4100	2570	5500	0.44
-241	.8750	1.5000	.875	1.500	0.7500	7/8-14	0.157	0.375	0.188	20.00	0.0023	1.10	900.0	5500	3390	7300	0.65
-261	.8750	1.6250	.875	1.500	0.7500	7/8-14	0.157	0.375	0.188	20.00	0.0023	1.10	900.0	5500	3650	7300	0.73
-281	1.0000	1.7500	1.000	1.750	0.8750	1-14	0.157	0.438	0.188	20.00	0.0023	1.30	1350.0	7700	4500	10300	1.02
-301	1.0000	1.8750	1.000	1.750	0.8750	1-14	0.157	0.438	0.188	20.00	0.0023	1.30	1350.0	7700	4828	10300	1.38
-321	1.1250	2.0000	1.250	2.000	1.0000	1 1/8-12	0.188	0.500	0.188	24.00	0.0023	1.45	1650.0	10400	6550	13800	1.59
-361	1.1250	2.2500	1.250	2.000	1.0000	1 1/8-12	0.188	0.500	0.188	24.00	0.0023	1.45	1650.0	10400	7370	13800	1.89
-401	1.2500	2.2500	1.500	2.250	1.1250	1 1/4-12	0.188	0.562	0.188	30.00	0.0023	1.70	2050.0	16200	4250	21600	2.78
-441	1.2500	2.5000	1.500	2.250	1.1250	1 1/4-12	0.188	0.562	0.188	30.00	0.0023	1.70	2050.0	16200	10200	21600	3.21
-481	1.5000	2.7500	1.750	2.500	1.2500	1 1/2-12	0.188	0.625	0.250	30.00	0.0023	2.10	3000.0	24600	13400	32800	4.58
-521	1.5000	3.0000	1.750	2.500	1.2500	1 1/2-12	0.188	0.625	0.250	30.00	0.0023	2.10	3000.0	24600	14500	32800	5.19
-561	1.7500	3.2500	2.000	2.750	1.3750	1 3/4-12	0.188	0.688	0.250	30.00	0.0023	2.44	3000.0	31200	18300	41700	7.13
-641	2.0000	3.5000	2.250	3.500	1.5000	2-12	0.188	0.750	0.250	30.00	0.0023	2.80	3000.0	41000	24100	54700	11.30

• OIL HOLE "F" DRILLED FROM THE FLANGE END OF THE STUD TO THE RADIAL OIL HOLE ONLY.

1/ INSTALLATION TORQUE LUBRICATED THREADS.

(B) DENOTES CHANGES

INCH - POUND

PREPARING ACTIVITY: NAVY-AS CUSTODIANS: ARMY-AT NAVY- AIR FORCE-99 DLA- REVIEW: ARMY-AT, NAVY-AS, AF-11,99, DLA-IS USER: PROJECT NUMBER: 3110-0852	MILITARY SPECIFICATION SHEET TITLE BEARING, ROLLER, NEEDLE, TRACK ROLLER, INTEGRAL HEAVY STUD, TYPE IX, ANTIFRICTION, INCH	SPECIFICATION SHEET NUMBER MS21441 05 DEC 94 REV B SUPERSEDING MS21441A 31 MAR 81 AMSC- N/A FSC 5310

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THE REQUIREMENTS FOR ACQUIRING THE PRODUCT(S) DESCRIBED HEREIN SHALL CONSIST OF THIS SPECIFICATION SHEET AND THE ISSUE OF THE FOLLOWING SPECIFICATION LISTED IN THAT SOLICITATION: MIL-P-3990

THIS SPECIFICATION IS APPROVED FOR USE BY ALL DEPARTMENTS AND AGENCIES OF THE DEPARTMENT OF DEFENSE

Form Approved
OMB No. 0704-0188

B. TABLE II. TOLERANCE LIMITS

Dimensions in inches

d Basic Stud Diameter		Allowable Deviation From d of Single Mean Dia. d_{mp}		D Basic Outer Ring Outside Diameter		Allowable Deviation From D of Single Mean Dia. D_{mp}		Allowable Deviation From Outer Ring Width	
Over	Incl	High	Low	Over	Incl	High	Low	High	Low
0.1250	1.5000	+0.0015	0.0000	0.4375	4.0000	+0.0020	-0.0005	+0.005	-0.005

REQUIREMENTS:

1. MATERIAL: Steel, MIL-S-8690, ASTM A304, A322, A331, QQ-S-700, Fed Std No. 66 Steel No. 50100, 51100, 52100. Seals and backing rings - acetal resin per L-P-392 or nylon per L-P-410, type 6/6 wear resistant grade elastomeric polyester. Construction optional.

2. FINISH OR PLATING:

   A. Outer Ring - outer ring chrome plated per QQ-C-320, Class 2. O.D. and O.D. corners .0004-.0007 in. thickness, faces minimum .0003 in. thickness.

   (B) B. Stud, all external surfaces, except for the unthreaded portion of stud shank, zinc-nickel in accordance with AMS 2417, Type 2, or cadmium plated in accordance QQ-P-416, Type II, class 2, with a minimum thickness of .0003 inches, maximum thickness of .0006 inches.

   (B) C. Endplates - all exposed surfaces zinc-nickel in accordance with AMS 2417, type 2, or cadmium plated in accordance with QQ-P-416, Type II, class 2, with a minimum thickness of .0003 inches, maximum thickness of .0006 inches.

3. LUBRICANT: MIL-G-81322 or MIL-G-23827. All bearings shall be prepacked with grease conforming to MIL-G-81322 unless otherwise specified. If MIL-G-23827 is required add the letter G after the MS part number.

4. MARKING: The marking shall consist of the MS part number and the manufacturer's identification in accordance withMIL-STD-130.

5. PART NUMBER: The MS part number shall consist of the MS number plus the dash number. Example: MS21441-081 or MS21441-081G.

6. (E) PACKING: Bearings shall be individually packaged to the requirements of MIL-P-197. Packaged and marked with manufacturer's name or trademark, and date of lubrication by month and year.

7. DIMENSIONS: All dimensions are after plating and are in inches.

8. TEMPERATURE: Bearing operating temperature range -65°F to 250°F.

9. Remove all burrs and sharp edges.

NOTES:

1. DESIGN FEATURES: For design feature purposes, this standard takes precedence over documents referenced herein.

2. REFERENCED DOCUMENTS: Shall be of the issue in effect on date of invitations for bid.

3. LIMIT LOAD: The limit load rating listed can be defined as the maximum radial load which can be applied to a bearing without impairing the subsequent function of the bearing. The ultimate or static fracture load rating is not less than 1.5 times the limit load rating. The capacity as a track roller is the load bearing will carry as a track roller for an L-10 life of 20,000 revoloutions. To fully realize limit load rating, compensation must be provided for stud deflection in order to assure full track contact under load.

PREPARING ACTIVITY: NAVY-AS CUSTODIANS: ARMY- AT NAVY- AIR FORCE- 99 DLA- REVIEW: ARMY-AT, NAVY-AS,AF-11,99, DLA-IS USER: PROJECT NUMBER: 3110-0852	MILITARY SPECIFICATION SHEET TITLE BEARING, ROLLER, NEEDLE, TRACK ROLLER, INTEGRAL HEAVY STUD, TYPE IX, ANTIFRICTION, INCH	SPECIFICATION SHEET NUMBER MS21441 05 DEC 94 REV B SUPERSEDING MS21441A 31 MAR 81 AMSC- N/A FSC 5310
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